Java

Case study –3

Bus Ticket Booking System

1 code

```java
import java.util.InputMismatchException;
import java.util.Scanner;

class InsufficientSeatsException extends Exception{  3 usages  new *
    InsufficientSeatsException(String message) {  1 usage  new *
        super(message);
    }
}


class Bus {  2 usages  new *

    int as;  5 usages

    Bus(int as) {  1 usage  new *
        this.as = as;
    }
    public void bookTicket(int age, int seats) throws InsufficientSeatsException {  1 usage  new *

        if (age <= 0) {
            throw new IllegalArgumentException("Age can't be less than 0.");
        }
        if (seats <= 0) {
            throw new IllegalArgumentException("Number should be greater than 0.");
        }
        if (seats > as) {
            throw new InsufficientSeatsException("Only " + as + " seats are available.");
        }


        as -= seats;

        System.out.println("Booking Successful!");
        System.out.println("Seats booked: " + seats);
        System.out.println("Remaining seats: " + as);
    }
```

```java
public class casestudy11 {  new *

    public static void main(String[] args) {  new *

        Scanner sc = new Scanner(System.in);
        Bus bus = new Bus( as: 10);

        try {

            System.out.print("passenger Age: ");
            int age = sc.nextInt();

            System.out.print("number of Seats: ");
            int seats = sc.nextInt();

            bus.bookTicket(age, seats);

        }
        catch (InputMismatchException e) {
            System.out.println("Invalid input! Please enter numbers only.");
        }


        catch (IllegalArgumentException e) {
            System.out.println(e.getMessage());
        }

        catch (InsufficientSeatsException e) {
            System.out.println(e.getMessage());
        }

        finally {
            sc.close();
        }
    }
}
```

2 output

```
passenger Age: 20
number of Seats: 2
Booking Successful!
Seats booked: 2
Remaining seats: 8

Process finished with exit code 0
```